Università degli Studi di Napoli
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Statistical Learning and Data Analysis – Module A

STATISTICAL TESTING

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Statistical Testing

- Statistical testing is an inferential procedure to verify the acceptance of a **null hypothesis** H_0 against an **alternative hypothesis** H_1 on the basis of the sample derived from a population.
  - ✓Parametric testing refers to hypothesis concerning a parameter of the model population, such as H_0: $\mu = \mu_0$ *against* H_1: $\mu > \mu_0$.
  - ✓Non-parametric testing refers to hypothesis concerning the probability model of the population, i.e., H_0: $f(x, \theta) \sim N(\mu, \sigma^2)$ *against* it is not true.
- The **statistical test** is a judgment of how coherent is the sample to be extracted from the population as defined by H_0 with respect to test bed represented by H_1.

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Key elements in statistical testing

- **Hypotheses**: H_0 *versus* H_1
- **Testing variable**: D_n is a random variable defined as a function of the n-dimensional random sample: $D_n = d(X_1, \dots, X_n)$
  - It is typically a useful transformation of the Estimator that makes possible the use of a known probability table. As an example, under H_0: $\mu = \mu_0$, assuming known variance and normal distribution (or large samples and unbiased variance estimate) the testing variable can be
$$D_n = \frac{\bar{X}_n - \mu_0}{\sigma/\sqrt{n}} \sim N(0,1)$$
  - d_n is the empirical value of the testing variable for a given sample.
- **Decision rule** allows to split the possible values of D_n into two regions:
  - ✓ $D_n \in A$, acceptance region A (samples coherent with H_0)
  - ✓ $D_n \in C$, rejection region C (samples coherent with H_1).

  such to optimize the risk to make a wrong decision.

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Decision or Payoff table

- Two are the possible Errors:
  - ✓ Reject the null hypothesis H_0 when this is true (Error Type I)
  - ✓ Accept the null hypothesis H_0

Decision / Population	Null hypothesis is accepted $D_n \in A$	Null hypothesis is rejected $D_n \in C$
H_0 is true	Right Decision	E_1: Error Type I
	$P(\bar{E}_1) = P(D_n \in A \mid H_0) = 1 - \alpha$	$P(E_1) = P(D_n \in C \mid H_0) = \alpha$
H_0 is NOT true	E_2: Error Type II	Right Decision
	$P(E_2) = P(D_n \in A \mid \bar{H}_0) = \beta$	$P(\bar{E}_2) = P(D_n \in C \mid \bar{H}_0) = 1 - \beta$

- The split of the possible values of the testing variable D_n into two regions A and C determines two risks: α and β. Enlarging A with respect to C yields to decrease α and increase β, and vice versa. *The two risks move in opposite way.* An optimization criterion needs to be defined.

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Decision rule

- The most sever error is the Error Type I, so that one should reduce to the minimum the risk α to a value such as 0.001 or 0.05 or 0.1. This is known **significance level**.
- **Lemma of Neyman and Pearson:** *find the region of rejection C for a given α minimizing β or, equivalently, maximizing the power of test $1-\beta$.* Such critical region is at the extreme values or tails of the sampling distribution of the testing variable.
- Alternatively, for a given sample, calculate $P(|D_n| > |d_n||H_0) = p$ where p is called p-value:
  - ✓ lower is the p-value higher the significance to reject the null hypothesis.
- The decision rule can be based on the p –value compared to the significance level α:
  - ✓ Accept H_0 if $p > \alpha$ *(the empirical value makes the sample to be coherent with H_0)*
  - ✓ Reject H_0 if $p \leq \alpha$ *(the empirical value makes the sample NOT to be coherent with H_0)*
- In the output, nearby a very low p-value it appears a number of "*" corresponding to alternative choices α = 0.001 (***), 0.05 (**), 0.1 (*).

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Null hypothesis – Alternative Hypothesis

simple

"If the probability distribution of the random variable under study is fully specified, these are examples of simple hypotheses:"
H_0: p = 0.6 (Bernoulli population)
H_1: μ = 177 (normal population with known σ^2)
H_0: λ = 3.2 (Poisson population)

composed

Composed hypothesis example:

$H_0 : p \leq 0{,}5;$
$H_1 : \mu > 175;$
$H_1 : \mu \neq 150.$

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Observed significance level

The observed significance level, which we denote by α_{oss}, is the probability that the test statistic takes a value greater than the one obtained from the observed sample
(a more extreme value), assuming that the null hypothesis is true.
In symbols:

$$\alpha_{oss} = P(Z_{\overline{X}} > z_{\overline{x}} \mid \mu = \mu_0) = 1 - \Phi(z_{\overline{x}})$$

value taken by the test statistic $Z_{\overline{x}}$ in the observed sample

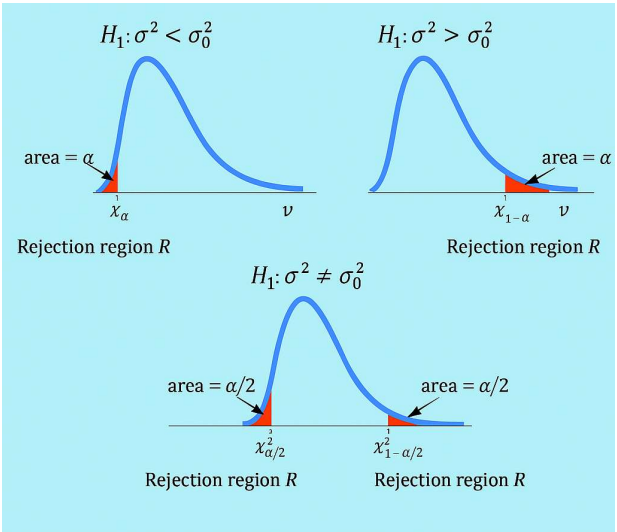
Given the hypothesis system

The null hypothesis is rejected if the observed significance level is less than α.

Hypothesis systems and rejection regions for testing the hypothesis about σ^2

Test:

$$V = \frac{(n-1)S^2}{\sigma_0^2}$$



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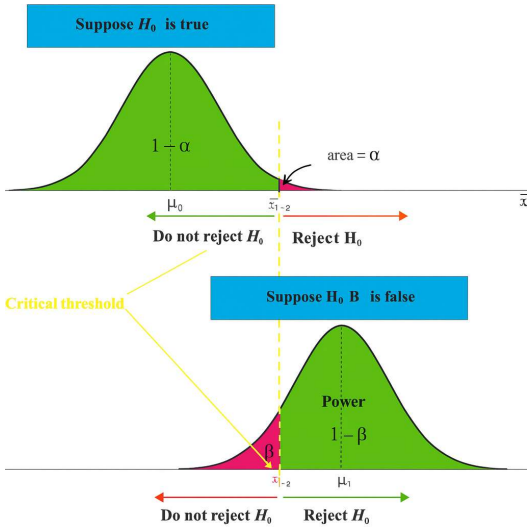
β probability and power of the test for the mean of a normal population

In the graph alongside, the probabilities of Type I and Type II errors are illustrated for testing the hypotheses $H_0 : \mu = \mu_0$, against $H_1 : \mu = \mu_1$, with μ_1 greater than μ_0.

$$\alpha = P(\bar{X} > \bar{x}_{1-\alpha} \mid \mu = \mu_0)$$

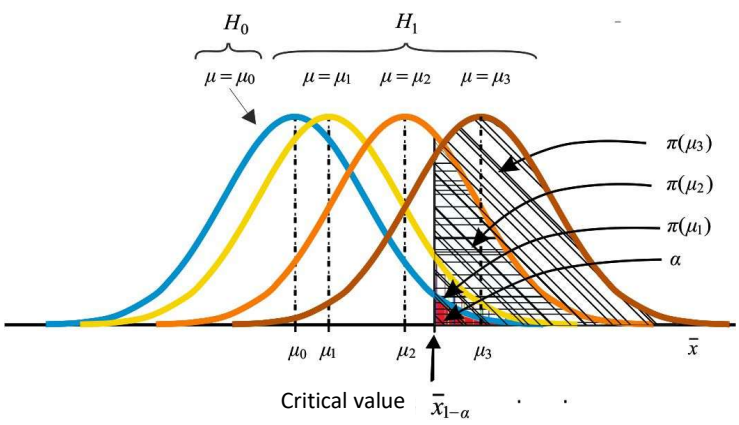
$$\beta = P(\bar{X} < \bar{x}_{1-\alpha} \mid \mu = \mu_1)$$



The probability β depends from μ_1

Power of the test on the mean and alternative hypothesis

The graph shows the power of the test on the mean (the area under the curve to the right of the critical value) as the mean under the alternative hypothesis increases.



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